Twi Voice Notes Transcriber System

Idea

The Twi Voice Notes Transcriber is a system designed to convert spoken Asante Twi audio into written text. The project aims to make it easier for users to capture, store, and reuse spoken information without manually transcribing voice recordings. By transforming voice notes into readable content, the system can improve communication, accessibility, and productivity for Twi speakers.

The solution can be useful for students, professionals, journalists, content creators, and anyone who frequently relies on voice recordings in their daily activities.

Mission

To preserve and enhance communication in Asante Twi by transforming spoken language into accessible text, fostering understanding and collaboration between native speakers and wider audiences.

Vision

To create a digitally connected society where local languages such as Twi can be easily understood, shared, and appreciated by people regardless of their linguistic background.

Features / Functionalities

- **Audio Upload:** Users can upload recorded audio files in supported formats such as .ogg, .mp3, or .wav.
- **Voice Recording:** The system allows users to record audio directly within the application.
- **Speech-to-Text Transcription:** The application converts spoken Asante Twi audio into readable text automatically.
- **Transcription Display:** Generated transcriptions are displayed clearly for users to read and review.

- **Copy and Download Options:** Users can copy the transcribed text or download it for future use.
- **Audio Playback:** Users can replay uploaded or recorded audio while reviewing the transcription.
- **Error Notifications:** The system provides feedback when unsupported file formats are uploaded, audio quality is poor, or transcription processing fails.
- **User-Friendly Interface:** The application is designed with a simple and intuitive interface for ease of use.

Possible Problems

- **Limited Training Data:** Asante Twi is a low-resource language, so obtaining large amounts of high-quality speech data may be difficult.
- **Accent and Dialect Variations:** Different pronunciations and dialects may affect transcription accuracy.
- **Background Noise:** Poor audio quality or noisy environments may reduce the system's performance.
- **Transcription Accuracy:** The system may struggle with slang, mixed languages, or unclear speech.
- **Processing Speed:** Large audio files may take longer to process and transcribe.

Possible Future Enhancements

- **Real-Time Transcription:** Enable live speech-to-text transcription during conversations or presentations.
- **Translation Support:** Add translation features between Twi and English.
- **Mobile Application:** Develop a mobile version of the system for easier accessibility.
- **Speaker Identification:** Allow the system to recognize and separate multiple speakers in a recording.
- **Subtitle Generation:** Generate subtitles for Twi videos and media content automatically.
- **Cloud Storage Integration:** Allow users to save and manage transcriptions online.
- **Support for More Ghanaian Languages:** Expand the system to support additional local languages such as Ga and Ewe.